

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Pharm Pharmaceutical Technology

University Theory Examination November / December 2015

PH819.04 Pharmaceutical Product Development

Date: 14/12/15, Monday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** **A)** What is Biopharmaceutical-classification system? Discuss it's significance in development of the dosage form. **05**
- B)** Discuss various applications of Complexation in product development. **05**
- Q.2** Discuss in brief about ICH Q1A(R2) and Q1D guidelines. **10**

OR

- Q.2** **A)** Explain steady state diffusion and discuss Fick's law of diffusion. **05**
- B)** Enlist the dissolution apparatus as per the USP along with their applications and write in brief about the USP dissolution apparatus 2. **05**
- Q.3** **Answer the following (Any three)** **15**
- (A)** Evaluation of stability in solid and disperse system
- (B)** Cyclodextrin as a complexing agent.
- (C)** Model Independent method of Dissolution profile comparison.
- (D)** Accelerated stability study

SECTION-II

- Q.4** 1. Discuss briefly various Pharmaceutical factors affecting preformulation studies. **05**
2. Prodrug formulation of drug is the preformulation study. **05**
- Q.5** Explain in detail
- a. Ion exchange polymer **05**
- b. Cyclodextrin **05**

OR

- Q.5** Justify the comments:
- a. Ethyl cellulose is used in the fast disintegrating tablet film coating. **2.5**
- b. Sodium starch glycolate can be used in controlled release formulation **2.5**
- c. Amphotericin B is more bioavailable. **2.5**
- d. PLGA is used in tablet formulation. **2.5**

Q.6 **Answer the following (Any three)**

- (A)** Write a note on directly compressible vehicles. **05**
- (B)** What are the selection parameters for any excipients for pharmaceutical dosage form? **05**
- (C)** What are the polymer merits in novel dosage form design? **05**
- (D)** Why is the preformulation study carried out before development of dosage form? **05**